

Math 196 Lecture Notes: Week 4

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April 2026

0.1 Review of Span and Linear Subspaces

0.1.1 Span

Given vectors $v_1, \dots, v_r \in \mathbb{R}^n$, their *span* is the set of all vectors that can be expressed as a linear combination

$$x_1 v_1 + \dots + x_r v_r$$

we denote this $\text{span}\{v_1, \dots, v_r\}$. Letting A be the matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix}$$

we have

$$A \begin{bmatrix} x_1 \\ \vdots \\ x_r \end{bmatrix} = x_1 v_1 + \dots + x_r v_r$$

and so

$$\text{span}\{v_1, \dots, v_r\} = \{Aw \mid w \in \mathbb{R}^r\}$$

Viewing A as a function, this is the set of outputs that the function can yield. In general, this is called the *image* of function. We write $\text{im}(A)$ to denote this.

Example 1. In $\mathbb{R}^3$, we have the following examples of spans.

1. $\text{span}\{e_1\}$ is the x -axis. In general, $\text{span}\{v\}$ is a line in the direction of v .
2. $\text{span}\{e_1, e_2\}$ is the xy -plane. Typically, the span of two vectors is a plane through the origin.
3. $\text{span}\{e_1, e_2, e_3\}$ is all of space.

0.1.2 Linear Subspaces

A linear subspace is a set of vectors V in $\mathbb{R}^n$ with the following properties:

1. $0 \in V$
2. If $v, w \in V$, then $v + w \in V$
3. If $v \in V$, $a \in \mathbb{R}$, $av \in V$.

For any collection of vectors. Their span is a linear subspace. In fact, any linear subspace in $\mathbb{R}^n$ is the span of $\leq n$ vectors. One can think of the definition of linear subspaces as an alternative characterization of spans. In $\mathbb{R}^3$, we will see that linear subspaces can be classified as follows:

1. $\{0\}$, the origin.
2. Lines through the origin.
3. Planes through the origin.
4. All of $\mathbb{R}^3$.

Today we will discuss the thing that makes this classification possible, which is the notion of dimension.

0.1.3 Image and Kernel

Given an $m \times n$ matrix A , representing a linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^m$, we can associate with it two subspaces:

1. The image $\text{im}(A)$. This is the set of vectors in $\mathbb{R}^m$ that can be written Av for some $v \in \mathbb{R}^n$. This is the set of outputs of the function, or its range. We saw that this is the span of the columns of A .
2. The kernel $\ker(A)$. This is the set of vectors v in $\mathbb{R}^n$ such that $Av = 0$. This can be viewed as the solution set to the linear system $Ax = 0$.

On Thursday, we saw that our method of solving linear systems gives us a way to express a span as the solution set of a system of equations, or the solution set of a system of equations as a span. These are two representations of linear subspaces, each of which is useful for different purposes.

Recall that a function $f : X \rightarrow Y$ is invertible if and only if it is:

1. surjective: for any $y \in Y$, there is an x with $f(x) = y$.
2. injective: for any $x_1, x_2 \in X$, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. If equivalently, if there is an x satisfying $f(x) = y$, it is unique.

The image and kernel determine if these properties are satisfied for a linear transformation. In particular, $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is surjective if $\text{im}(T) = \mathbb{R}^m$, by definition. I claim that T is injective if and only if $\ker(T) = \{0\}$.

Proof. First of all, if T is injective, then 0 must be the unique solution to $T(x) = 0$, so $\ker(T) = \{0\}$. Now we consider the converse direction. Suppose that $\ker(T) = \{0\}$. Suppose that $T(x_1) = T(x_2)$. We need to show that $x_1 = x_2$. We have that

$$T(x_1 - x_2) = T(x_1) - T(x_2) = 0$$

so $x_1 - x_2 \in \ker(T)$. By assumption, this means that $x_1 - x_2 = 0$, so $x_1 = x_2$. □

Thus the image and kernel of a linear transformation tell us something about the existence and uniqueness of solutions to linear systems.

0.2 Plan for the week

This week we will introduce linear dependence, bases and dimension of subspaces. This will be the last piece of the technical core of linear algebra that forms the first part of the course, and will allow us to make sense of a lot of the things we have been seeing up to this point. After this, we will return to the dot product, discussing its geometric significance in higher dimensions. This will start the second part of the course, which will lead us to least squares approximation and linear regression.

1 Linear Dependence and Bases

1.1 Linear Dependence

When we describe a linear subspace by giving a spanning set, some of the vectors may be redundant. For example, if we consider

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\}$$

we have

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x + 4y \end{bmatrix} = (x + 2y) \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Therefore, this subspace can also be represented as just

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Geometrically, both vectors lie on the same line [draw picture]. In this case, this happens because

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Another way of saying this is that these vectors satisfy the relation

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 0$$

Proposition. *Given a collection of vectors $v_1, \dots, v_r \in \mathbb{R}^n$, the following are equivalent.*

1. $\text{span}\{v_1, \dots, v_r\}$ can also be written as $\text{span}\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_r\}$, omitting some v_i .
2. Some v_i is in the span of the other vectors.
3. There is a linear relation

$$x_1 v_1 + \dots + x_n v_n = 0$$

where some $x_i \neq 0$.

Proof. To see that (1) and (2) are equivalent, we notice that if v_i can be omitted, then it has to be in the span of the other vectors, because $v_i \in \text{span}\{v_1, \dots, v_r\}$. Conversely, if v_i is in the span of the other vectors, say

$$v_i = \sum_{j \neq i} a_j v_j$$

Then in any linear combination

$$x_1 v_1 + \dots + x_i v_i + \dots + x_n v_n$$

we can replace v_i ,

$$x_1 v_1 + \dots + x_i \left(\sum_{j \neq i} a_j v_j \right) + \dots + x_n v_n$$

which comes out to a linear combination of the remaining vectors. Thus v_i is redundant.

To see that (2) and (3) are equivalent, if v_i is a linear combination of the other vectors, we can rearrange this to get an equation like in (3). Conversely, if (3) holds with $x_i \neq 0$, then we can solve for v_i , expressing it as a linear combination of the other vectors. $\square$

If any/all of these conditions hold, we say that the vectors $v_1, \dots, v_r$ are *linearly dependent*. If not, we say that they are *linearly independent*.

Example 2. The vectors

$$u = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, w = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

in $\mathbb{R}^3$ are linearly dependent, as $u + v - w = 0$. Notice that these three vectors lie in the xy -plane. The vectors u, v are linearly independent. If

$$xu + yv = \begin{bmatrix} x + y \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

then $y = 0$, and so $x = 0$.

An important feature of linearly independent collections of vectors is that any element of their span has a *unique* representation as a linear combination. Indeed, if

$$x_1 v_1 + \dots + x_r v_r = y_1 v_1 + \dots + y_r v_r$$

then

$$(x_1 - y_1)v_1 + \dots + (x_r - y_r)v_r = 0$$

and so each $x_i - y_i = 0$, that is, $x_i = y_i$.

Conclusion. For vectors $v_1, \dots, v_r$, if we put them into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix}$$

Then $v_1, \dots, v_r$ are linearly independent if and only if A is injective, if and only if $\ker(A) = \{0\}$.

To see this, notice that linear relations $x_1 v_1 + \dots + x_r v_r$ correspond to vectors

$$\begin{bmatrix} x_1 \\ \vdots \\ x_r \end{bmatrix} \in \ker(A)$$

The kernel being 0 means that there are no nontrivial relations. The matrix being injective is exactly the uniqueness remarked above.

1.2 Bases

If $V = \text{span}\{v_1, \dots, v_r\}$ is a subspace and $v_1, \dots, v_r$ are linearly independent, then we say that $v_1, \dots, v_r$ form a *basis* of V . In this case, any element of V can be expressed in exactly one way as a linear combination of $v_1, \dots, v_r$. In particular, if $\text{span}\{v_1, \dots, v_r\} = \mathbb{R}^n$ and $v_1, \dots, v_r$ are linearly independent, we say that these vectors form a basis for $\mathbb{R}^n$. (We will see that in this case it must be that $r = n$). We have already been working with a basis for $\mathbb{R}^n$, consisting of the vectors $e_1, \dots, e_n$. This is called the *standard basis*.

Many other bases exist. In fact:

Proposition. Vectors $v_1, \dots, v_n \in \mathbb{R}^n$ form a basis if and only if the $n \times n$ matrix

$$\begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

is invertible.

Proof. This is a translation of the fact that a matrix is invertible if and only if it is injective, meaning that the columns are linearly independent, and surjective, meaning that the span of the columns is all of $\mathbb{R}^n$. $\square$

Therefore, the vectors

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

form a basis for $\mathbb{R}^2$, for example. You saw another basis for $\mathbb{R}^2$ on the first homework. If we are given a spanning set for a subspace

$$V = \text{span}\{v_1, \dots, v_r\}$$

then we can convert it into a basis using the following procedure. If it is not already a basis, then one vector v_i must be redundant, it can be written as a linear combination of the others. Remove this one. Proceed until no vectors are redundant.

This is simple to describe, but to apply it we need to be able to decide if a given vector is redundant. This requires us to solve a linear system to determine if the vector is in the span of the others, so if we need to do this many times, this can become computationally intense. There is an easier method, which allows us to find a basis by applying Gaussian elimination a single time.

Given vectors $v_1, \dots, v_r$, we put them into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix}$$

We then bring this to reduced row echelon form:

$$B = \begin{bmatrix} \vdots & & \vdots \\ w_1 & \dots & w_n \\ \vdots & & \vdots \end{bmatrix}$$

The first thing to observe is that the pivot columns of B form a basis for $\text{span}\{w_1, \dots, w_n\}$. Indeed, this span is exactly the collection of vectors whose only nonzero entries are in the position of nonzero rows of B . We will argue that the corresponding columns of A form a basis of its image. The point is that the kernel of A and the kernel of B are the same. The kernels are the solution sets of $Ax = 0$, $Bx = 0$ respectively, so this is to say that row operations do not change the solution set of a linear system.

Because the kernel records all linear relations between the columns, this implies that the columns of A which become pivot columns are linearly independent, and that any additional column can be written as a linear combination of them.

Example 3. Find a basis for the span of the following vectors in $\mathbb{R}^3$:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, v_4 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

We package the vectors into a matrix and bring to reduced row echelon form:

$$\begin{bmatrix} 1 & -1 & 3 & 1 \\ 1 & 1 & -1 & 2 \\ 2 & 0 & 2 & 3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 3 & 4 \\ 0 & 2 & -4 & 1 \\ 0 & 2 & -4 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 3 & 4 \\ 0 & 2 & -4 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 3 & 4 \\ 0 & 1 & -2 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 9/2 \\ 0 & 1 & -2 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here the first and second columns are pivots. This tells us that v_1, v_2 form a basis for the subspace. These four vectors all lie in the plane spanned by v_1 and v_2 .

1.3 Dimension and Rank

From what we know so far, it is conceivable that we could find one basis for a subspace of $\mathbb{R}^n$ consisting of 2 vectors, and another consisting of 3 vectors, say. This would in effect give us two coordinate systems on the subspace, one where we use 2 coordinates, and other where we use 3. In this hypothetical world, it would not make sense to say that we have 2 degrees of freedom when choosing an element of the subspace, because we could just as well say that we have 3.

Thankfully, this does not happen, and the notion of “number of degrees of freedom” not only makes sense, it gives us a useful way of describing the size of vector subspaces.

Theorem 1. *For a subspace $V \subset \mathbb{R}^n$, any two bases of V have the same size.*

Proof Sketch. This comes down to the following statement. If $v_1, \dots, v_r$ are linearly independent elements of V , and $w_1, \dots, w_s$ span V , then $r \leq s$. If both the v_i and the w_j are bases, then we can apply this in both directions to get $r = s$. To prove the claim, we can write

$$v_i = \sum_j a_{ij} w_j$$

Let A be the matrix with entries a_{ij} , so

$$\begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix} = \begin{bmatrix} \vdots & & \vdots \\ w_1 & \dots & w_s \\ \vdots & & \vdots \end{bmatrix} A$$

If $r > s$, then this matrix has more columns than rows, so $Ax = 0$ must have a nonzero solution; it represents a system with more variables than equations. This means that there are $x_1, \dots, x_r$, not all 0, such that

$$\sum_i a_{ij} x_i = 0$$

for each fixed j . (This is an element of $\ker(A)$), But this means that

$$\sum_i x_i v_i = \sum_i \sum_j x_i a_{ij} w_j = \sum_j \left[\sum_i x_i a_{ij} \right] w_j = 0$$

(In other words, it is an element of the kernel of the matrix with columns v_i .) This contradicts the linear independence of the v_i . $\square$

Knowing this, we define the *dimension* of a subspace V to be the size of any basis. In particular, $\mathbb{R}^n$ has dimension n , as we expect. This gives us a way to see that an invertible matrix must be square. If a matrix is invertible and has n rows, then its columns form a basis for $\mathbb{R}^n$, so there must be n of them.

If A is a matrix, then $\text{im}(A)$ is a subspace. We define the rank of A to be $\text{rk}(A) = \dim(\text{im}(A))$. Above, we introduced a method of computing a basis for $\text{im}(A)$, taking columns corresponding to pivot columns in the reduced row echelon form.

Conclusion. $\text{rk}(A)$ is the number of pivot columns in the reduced row echelon form of A .

This gives us an effective way to compute the dimension of subspaces in terms of a spanning set.

Example 4. Find the dimension of

$$\text{span} \left\{ \begin{bmatrix} -2 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix} \right\}$$

The row operations go as follows:

$$\begin{bmatrix} -2 & 3 & 1 \\ 2 & 3 & 1 \\ 6 & -9 & -3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -2 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -3/2 & -1/2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We have a single pivot column, so the dimension is 1. These three vectors all lie on a line.

Previously, given a matrix, we saw how to write down a spanning set for its kernel. To do this, we solve the linear system $Ax = 0$ using Gaussian elimination. In reduced row echelon form, each non-pivot column gives us a free variable, and these let us write any element of the kernel as a linear combination of certain vectors. It turns out that these vectors are always linearly independent, and so form a basis for the kernel.

The reason for this is best illustrated with an example. Let's look at a matrix which is already in reduced row echelon form.

$$A = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If we translate $Av = 0$ into a linear system in variables w, x, y, z , w and y are pivot variables, and x and z are non pivots. We can solve for w and y :

$$w = 3x - 2z$$

$$y = 2z$$

and hence the general solution to $Av = 0$ is

$$\begin{bmatrix} 3x - 2z \\ x \\ 2z \\ z \end{bmatrix} = x \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

In this case we can see that the two vectors are linearly independent by looking at the second and fourth entries. If the linear combination is to be 0, then we must have $x = 0$, because the second entry is just x , and $z = 0$, because the fourth entry is just z . We can conclude from this that the kernel of this matrix has dimension 2, it is a plane in $\mathbb{R}^4$.

Because of the way we assign the free variables to non-pivot columns, this always happens, for each variable, there is an entry in the vector where it appears alone. We can conclude from this that $\dim(\ker(A))$ is the number of non-pivot columns in the reduced row echelon form of A . Because $\dim(\text{im}(A))$ is the number of pivot columns, we get the following.

Theorem 2 (Rank Nullity Theorem). *If A is an $m \times n$ matrix, representing a transformation $\mathbb{R}^n \rightarrow \mathbb{R}^m$, then*

$$\dim(\text{im}(A)) + \dim(\ker(A)) = n$$

The reason for the name is that $\dim(\text{im}(A))$ is the rank of A , and $\dim(\ker(A))$ is sometimes called the nullity.

2 Bases as Coordinate Systems

2.1 Coordinates

Given a basis $v_1, \dots, v_n$ for $\mathbb{R}^n$, we can think of this as giving us a way of addressing vectors. Indeed, for any vector $v \in \mathbb{R}^n$, we can write

$$v = x_1 v_1 + \dots + x_n v_n$$

for unique $x_1, \dots, x_n$, and we can view the numbers $x_1, \dots, x_n$ as coordinates for v . We may as well view these numbers as forming a vector in themselves: given a basis $\mathfrak{B} = (v_1, \dots, v_n)$,

$$[v]_{\mathfrak{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Example 5. In the basis $\mathfrak{B}$:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

for $\mathbb{R}^2$, find the coordinates for the vector

$$v = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

.

[Draw picture with new coordinate grid on the plane coming from this basis.]

We want to express v as a linear combination of the basis vectors, so we solve the linear system with augmented matrix

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 2 & 1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & -3 & 2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -2/3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 1/3 \\ 0 & 1 & -2/3 \end{array} \right]$$

Thus

$$v = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

and so

$$[v]_{\mathfrak{B}} = \begin{bmatrix} 1/3 \\ -2/3 \end{bmatrix}$$

If we think about what we are doing here, what we just computed is

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

In general, given a basis $\mathfrak{B} = (v_1, \dots, v_n)$, if we let

$$S = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

then we have the formula

$$[v]_{\mathfrak{B}} = S^{-1}v$$

In particular, the operation of translating a vector into its coordinates in a different basis is linear. Notice that if our basis is the standard basis $\mathfrak{S} = (e_1, \dots, e_n)$, then the matrix is I_n . The components of a vector are already its coordinates with respect to the standard basis.

The linearity of the change of basis operation tells us that most of vector arithmetic does not depend on the choice of basis:

$$\begin{aligned}[v + w]_{\mathfrak{B}} &= [v]_{\mathfrak{B}} + [w]_{\mathfrak{B}} \\ [av]_{\mathfrak{B}} &= a[v]_{\mathfrak{B}}\end{aligned}$$

The one exception is the dot product, our formula for it depends on the choice of basis.

2.1.1 Linear Transformations in Different Coordinate Systems

Not only can we find representations of vectors in the coordinate system given by a basis, we can also find representations of linear maps. Suppose that $v_1, \dots, v_n$ is a basis of $\mathbb{R}^n$, and $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation. We know how to find a matrix representation of T with respect to the standard basis, but we can also find one with respect to this new basis.

For each i , we consider the representation of $T(v_i)$ in the coordinate system given by the basis $\mathfrak{B} = (v_1, \dots, v_n)$, $[T(v_i)]_{\mathfrak{B}}$. These vectors form the columns of an $n \times n$ matrix C . This matrix is a representation of T in the following way. Given any vector v , we can compute $T(v)$ by first taking the coordinate vector $[v]_{\mathfrak{B}}$, applying C to it to get a new coordinate vector

$$C[v]_{\mathfrak{B}} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

and then translating these coordinates into the vector they represent,

$$T(v) = y_1 v_1 + \dots + y_n v_n$$

Thus C allows us to compute Tv while staying in the $\mathfrak{B}$ coordinate system. Another way to state this is that the matrix C satisfies the equation

$$[T(v)]_{\mathfrak{B}} = C[v]_{\mathfrak{B}}$$

for any vector v .

Example 6. Let T be the linear transformation of $\mathbb{R}^2$ given by a quarter turn, rotation through $\pi/2$. Find the matrix representing T in the basis $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

We need to find the coordinates of

$$T(v_1) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad T(v_2) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

with respect to the basis. This is equivalent to applying the matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1/3 & -2/3 \\ -2/3 & 1/3 \end{bmatrix}$$

(We computed this inverse last week.) We compute

$$\begin{aligned}[T(v_1)]_{\mathfrak{B}} &= \begin{bmatrix} 1/3 & -2/3 \\ -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -4/3 \\ 5/3 \end{bmatrix} \\ [T(v_2)]_{\mathfrak{B}} &= \begin{bmatrix} 1/3 & -2/3 \\ -2/3 & 1/3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -5/3 \\ 4/3 \end{bmatrix}\end{aligned}$$

Thus the matrix we want is

$$\begin{bmatrix} -4/3 & -5/3 \\ 5/3 & 4/3 \end{bmatrix}$$

If A is the matrix representation of T in the standard basis, we can turn the above process into a formula for the representation of T in a new basis. Again let S be the matrix whose columns are the new basis vectors $v_1, \dots, v_n$. Then the columns of C are obtained by applying T to the columns of S , then applying S^{-1} . In other words, $C = S^{-1}AS$. We can also see this using the equation

$$[T(v)]_{\mathfrak{B}} = C[v]_{\mathfrak{B}}$$

This comes out to

$$S^{-1}AS = C$$

and so

$$S^{-1}AS = C$$

Matrices A, C with $C = S^{-1}AS$, equivalently $A = SCS^{-1}$ are called *similar*. We can say now that this means they represent the same transformation with respect to different choices of basis.

Given a transformation T , it is often easier to answer questions about it after choosing a coordinate system so that the associated matrix representation is as simple as possible. This is the idea behind the last main topic of the course, diagonalization.

Example 7. Let T be the transformation given by reflection across the line $y = -x$. On the first homework, we saw that this is given by the matrix

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Consider the basis for $\mathbb{R}^2$ consisting of the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

We then have $T(v_1) = -v_1$, $T(v_2) = v_2$, so in this coordinate system, T is represented by the matrix

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Any reflection matrix is similar to this one, for a similar reason.

2.2 Abstract Vector Spaces

To apply tools from linear algebra, we often consider objects which are not themselves vectors, but which can be represented as vectors. You saw an example of this on the first homework, with a polynomial represented as a vector of coefficients. With a generalization of the notion of a linear subspace, and our view of bases as coordinate systems, we can build a convenient language to talk about this. We will not use this formalism much in this course, but I do think it is worth seeing, because it underlies much of the applicability of linear algebra to myriad problems in mathematics itself, such as the study of differential equations.

Definition. A *vector space* is a set V equipped with two operations:

1. Addition: given $v, w \in V$, we can define $v + w \in V$.
2. Scalar Multiplication: given $v \in V$ and $k \in \mathbb{R}$, we can define $kv \in V$.

We also ask for an element $0 \in V$. These operations should satisfy the rules of vector algebra that we know from $\mathbb{R}^n$, in particular:

$$\begin{aligned} a(v + w) &= av + aw, (a + b)v = av + bv \\ a(bv) &= (ab)v \\ a \cdot 0 &= 0, v + 0 = v \end{aligned}$$

In any vector space, the notions of linear combinations, span, linear subspace, linear independence, bases, and dimension all make sense, as we have defined them for subspaces of $\mathbb{R}^n$.

Example 8. Any subspace $V \subset \mathbb{R}^n$ is a vector space.

Example 9. Let P be the collection of polynomials in one variable t . We can add polynomials, and multiply them by constants, so P is a vector space. Let P_3 be the collection of all polynomials of degree at most 3. This contains 0, and is closed under addition and scalar multiplication, so is a subspace of P . Any polynomial of degree at most 3 can be uniquely written as a linear combination of $1, t, t^2, t^3$:

$$f(t) = a_3t^3 + a_2t^2 + a_1t + a_0$$

so this is a basis for P_3 , showing that $\dim(P_3) = 4$. The vector space P has infinite dimension; a basis for it is the infinite set $\{1, t, t^2, t^3, t^4, \dots\}$.

If we have a (finite) basis $u_1, \dots, u_n$ for V a vector space, we can put a coordinate system on V , as we did with subspaces before. In particular, given $v \in V$

$$[v]_{\mathfrak{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

where $v = x_1u_1 + \dots + x_nu_n$.

Example 10. With respect to the basis $\mathfrak{B} = (1, t, t^2, t^3)$, the polynomial $f(t) = t^3 - 2t - 1$ has coordinates

$$[f]_{\mathfrak{B}} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

2.2.1 Linear Maps Between Abstract Vector Spaces

The notion of a linear transformation also makes sense for abstract vector spaces. Letting V and W be vector spaces, a function $f : V \rightarrow W$ is linear if

1. $f(v_1 + v_2) = f(v_1) + f(v_2)$
2. $f(k \cdot v) = k \cdot f(v)$

An invertible linear function is called an *isomorphism* of vector spaces. If there is an invertible linear function $V \rightarrow W$, we say that V and W are *isomorphic*.

Remark. The word *isomorphic* means essentially “same shape.” One can think of two isomorphic vector spaces as having the same structure, with different names for the objects. The isomorphism f translates an object in V into an object in W .

Example 11. Any choice of basis $v_1, \dots, v_n$ for V yields an isomorphism

$$F : \mathbb{R}^n \rightarrow V$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \mapsto x_1v_1 + \dots + x_nv_n$$

Specifically, we have an isomorphism

$$F : \mathbb{R}^4 \rightarrow P_3$$

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto x_3 t^3 + x_2 t^2 + x_1 t + x_0$$

If $F : \mathbb{R}^n \rightarrow V$ is an isomorphism coming from a basis $\mathfrak{B}$, then $[v]_{\mathfrak{B}} = F^{-1}v$.

Any isomorphism between two vector spaces takes a basis of one to a basis of the other. This means in particular that $\dim V = \dim W$. This gives us another way of seeing why invertible matrices are always square.

Given bases $\mathfrak{B} = (v_1, \dots, v_n)$ for V , $\mathfrak{C} = (w_1, \dots, w_m)$ for W , we can represent a linear function by a matrix in the following sense. We want a matrix A so that

$$[T(v)]_{\mathfrak{C}} = A[v]_{\mathfrak{B}}$$

We can compute this matrix in much the same way as before. For each v_i , we compute $[T(v_i)]_{\mathfrak{C}}$. These vectors then form the columns of our matrix. If $V = W$ and we use the same basis $\mathfrak{B}$ on both sides, then this is completely analagous to the earlier construction.

Example 12. Consider the map $D : P_3 \rightarrow P_3$ taking a polynomial $f(t)$ to its derivative $f'(t)$. This is a linear map. To compute its matrix representation in the basis $1, t, t^2, t^3$, we compute

$$D(1) = 0$$

$$D(t) = 1$$

$$D(t^2) = 2t$$

$$D(t^3) = 3t^2$$

This give us a matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If we use a different basis for P_3 , we get a different matrix representation. In particular, we could choose instead $1, t, t^2/2, t^3/6$, in which case we get the matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

From this matrix representation, we can see that $\text{rk}(D) = 3$, $\dim(\ker(D)) = 1$.